

CONCEPT PROPOSAL

LG Energy Solution Self-Driving Laboratory

System Engineering · Orchestration · STARS Enterprise

Taehwa Lee



From Lab Automation to a Self-Driving Laboratory

SDL connects experiment design, execution, measurement, data, and decisions in one closed loop — not only individual equipment automation.

CURRENT

People connect each step

Experiment Design

Manual

Pre-processing · Sample Preparation

Manual

Measurement

Manual

Data Collection

Manual

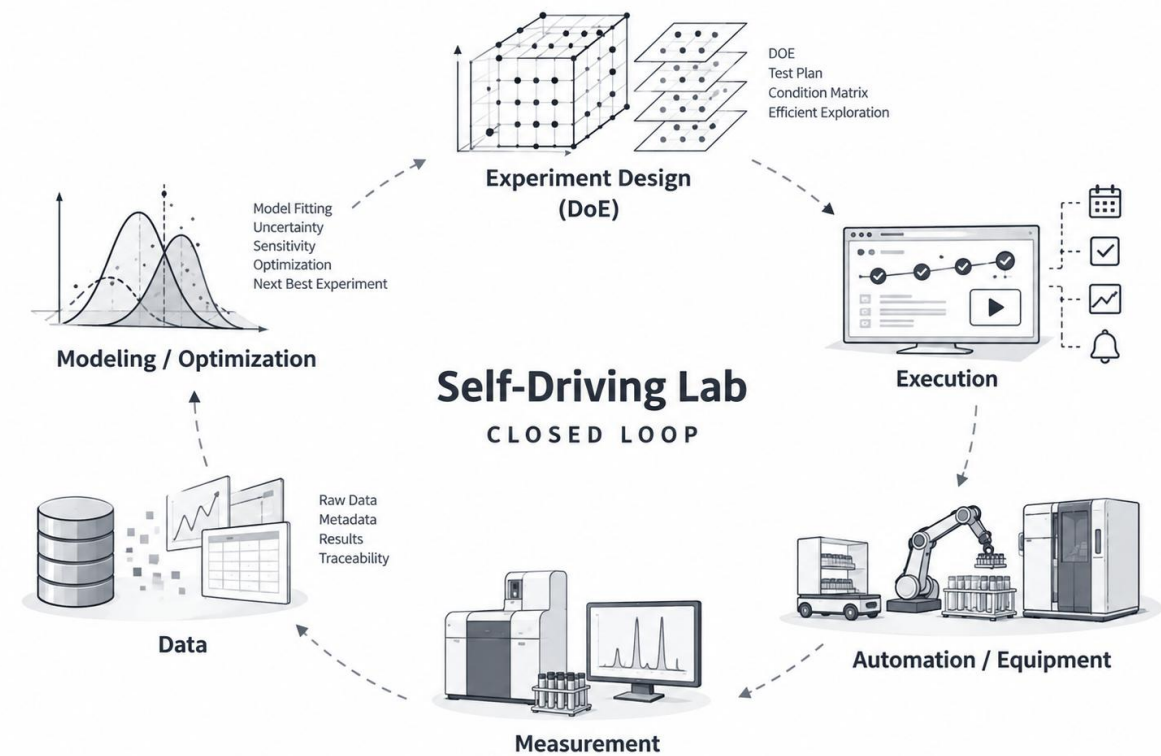
Analysis

Manual

Next Experiment Decision

Manual

TARGET · SELF-DRIVING LAB

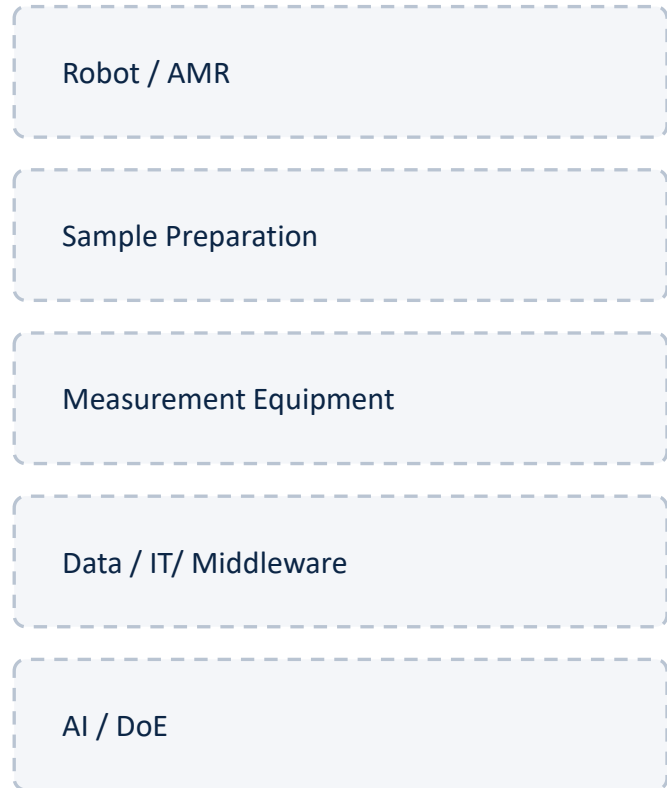


Trial & Error → Data-driven Closed-loop Experimentation

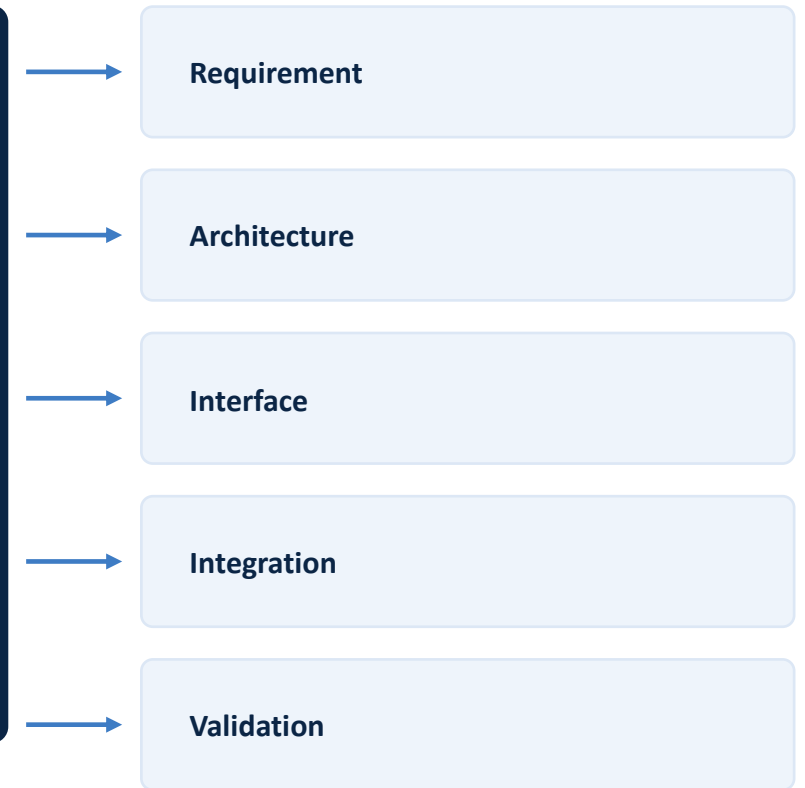
Who Connects the Automation?

If each technology is introduced separately, automation islands remain. The missing layer is System Engineering

TECHNOLOGY ISLANDS



ENGINEERING OUTPUT



“Who designs the laboratory as ONE SYSTEM?” — Define the whole laboratory first, before selecting each technology.

HORIBA Smart Lab Integration Capability

We combine physical laboratory integration experience and digital orchestration capability in one system design.

PHYSICAL & ENGINEERING

01 System Engineering

One design process from requirements to architecture, interface, and validation

Reference-based

02 Multi-Vendor Integration

Define control and connection scope for HORIBA, third-party, and customer equipment

Reference-based

03 Laboratory Automation

Design automated transport, sample handling, and equipment handover

Reference-based

DIGITAL

04 Workflow & Orchestration

Execution Planning · Control & Monitoring · Device Communication

STARS Enterprise

05 Data & Traceability

Metadata · Raw / Processed Data · Link experiment conditions and measurement results

Reference + STARS

06 Experiment Optimization

DoE · Modeling · Multi-objective Optimization

STARS Automation and Further collaboration with Academy

In this proposal: HORIBA is not only an analyzer supplier → HORIBA acts as a System Integration Partner.

Reference System Composition — Actual Proposal Scope

This page shows the actual proposed system scope, with customer information removed.

SYSTEM SCOPE · 3 Subsystems

01

Automated Transport

Autonomous transport · Equipment handover
Marker-based vision position correction

02

Data Collection & Orchestration

PC server · Orchestration
Link conditions and measurement data

03

Process & Evaluation Equipment

Catalyst Calciner · Coating · Drying evaluation
(Designed and manufactured by HORIBA)

INTEGRATION SCOPE · Responsibility by Equipment

MANUFACTURE

Catalyst Calciner · Coating equipment · Drying evaluation equipment
(In-situ optical monitoring integrated)

Includes sequence control · PLC · gas / lift / temperature control

PURCHASE

AMR · Robot Arm · Server

Standard products integrated into the architecture with a defined control scope

CONTROL / INTERFACE

1 sample-preparation unit + 8 measurement systems
(4 liquid measurement / 4 carrier measurement)

Existing customer and third-party equipment standardized for data integration

5

Functional Units

13

Equipment Types Integrated

26

Requirements Traced

5

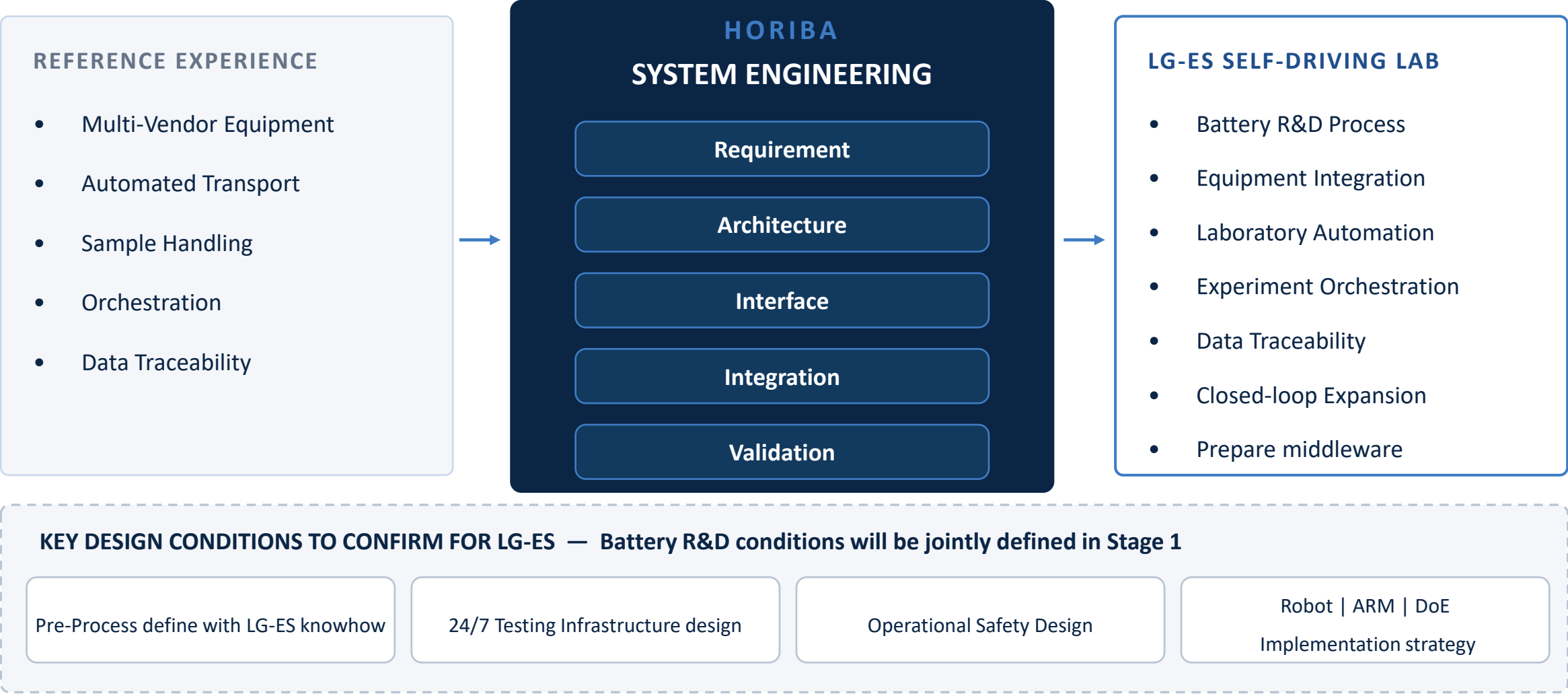
Defined Use Cases

This reference is based on a quotation/specification proposal to proven HORIBA Capability. It does not mean a completed or operating system.

Customer name, research field, equipment manufacturer/model, dimensions, and process conditions are confidential.

Apply the Reference Experience to LG-ES Battery R&D

We do not copy the reference laboratory. We apply the proven System Engineering & Integration methodology.



Clearly separate the reference from Battery R&D

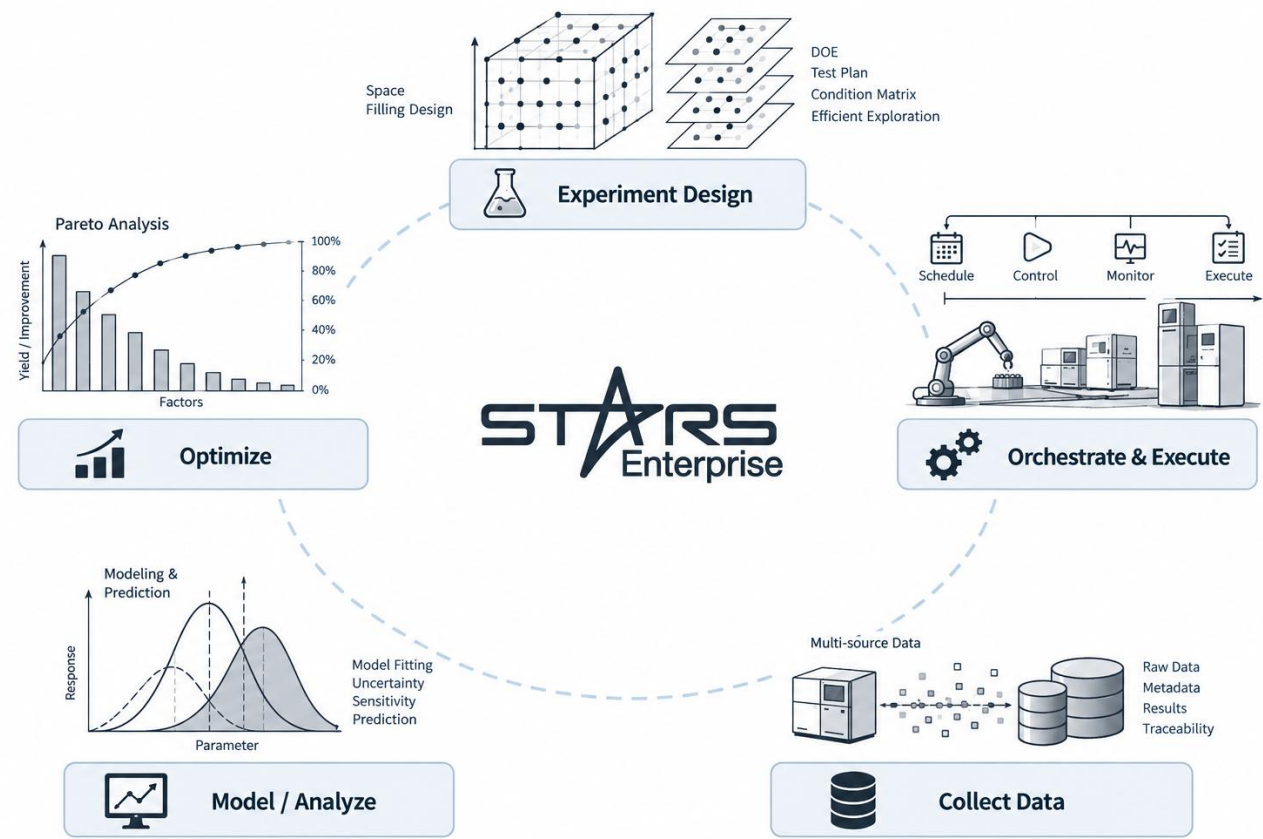
One Architecture for Physical Laboratory and Digital Intelligence

One architecture with six layers and a Data Spine supports both scalability and multi-vendor sourcing.



Design → Execute → Learn → Optimize

A digital backbone that connects experiment design, execution, data, and optimization in one loop.



ORCHESTRATION

Execution Planning · Control & Monitoring · Device Communication

DATA

Metadata · Raw Data · Processed Data · Data Utilization

UTILIZATION

DoE (factors · constraints) · Modeling (Kriging / Gaussian) · Multi-objective Optimization · Visualization

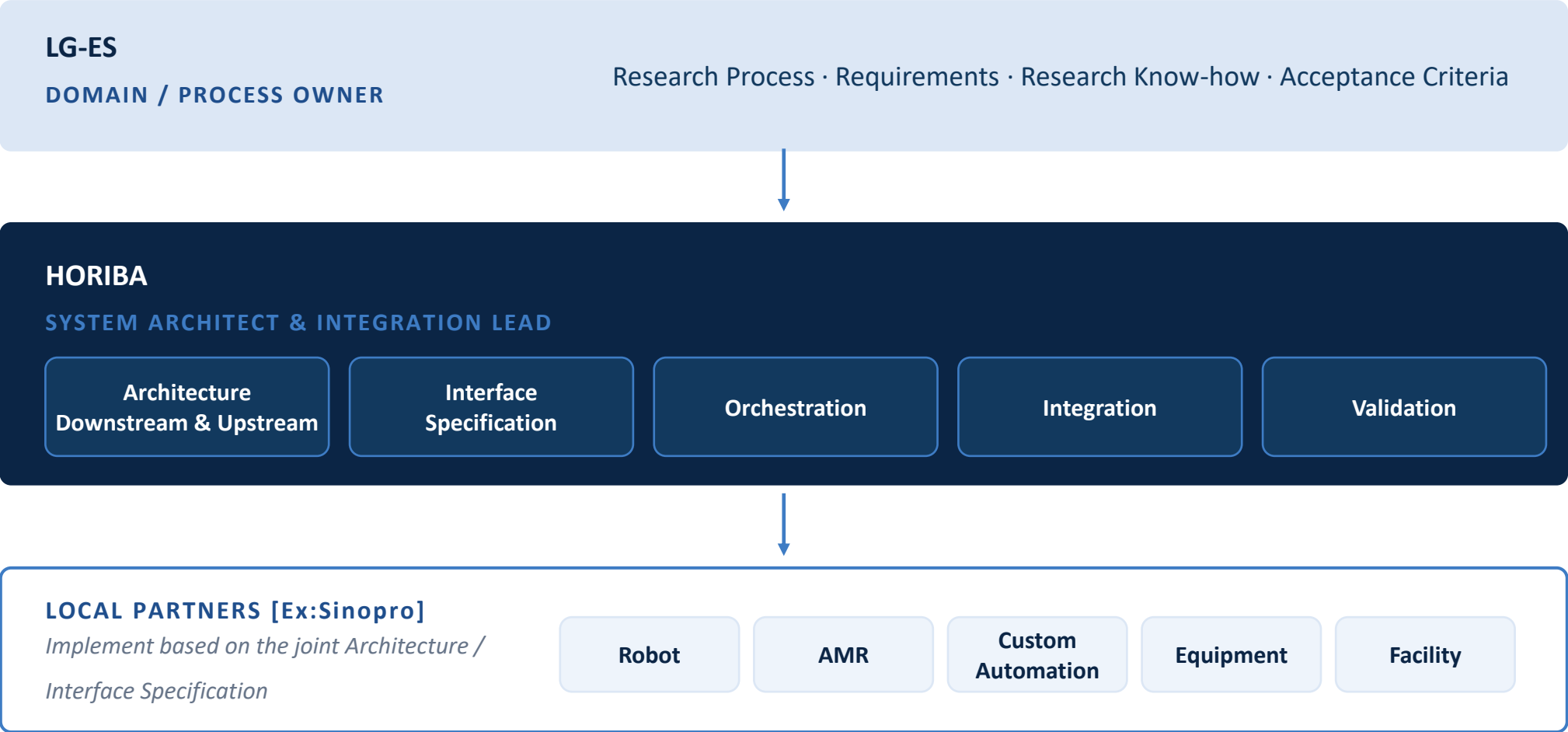
FUTURE

AI-driven Experiment Recommendation (Future Expansion)

One Architecture. Multiple Execution Partners.

Local partners do not need to create a fragmented system.

One architecture and interface specification enable effective local implementation.



Local Execution ≠ Fragmented Architecture

Define First. Integrate Second. Automate Third.

The first engagement defines the architecture and technical specification based on System Engineering.



The detailed scope and schedule will be agreed together in the next consulting stage.

Omoshiro-okashiku
Joy and Fun

おもしろ可笑

THANK YOU

Terima kasih 谢谢
Gracias
Tack ska du ha
Danke
Grazie
Σας ευχαριστώ πάρα πολύ
धन्यवाद
شُكْرًا
ขอบคุณครับ
Большое спасибо
Obbrigado
Cảm ơn
Dziękuję
Merci
감사합니다